SOFTWARE ENGINEERING

LAB:7 FUZZING FOR INPUT VALIDATION BUGS

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SEC:F

SCREENSHOTS:

SS1-ORIGINAL PROGRAM OUTPUT:

```
PS D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7> py -3.11 daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler:25
Age Group and Hours: ('Toddler', 25.0)

==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 25
Average Hours: 25

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3000|250
Payable Fee: 3290.0
PS D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7>
```

SS2-FAILING HYPOTHESIS TESTS:

```
PS D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7> py -3.11 -m pytest test_daycare_fees.py
    |   File "D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 28, in test_payable_fee_no_crash
    |     assert False, f"payable_fee crashed with: {e!r}"
    | AssertionError: payable_fee crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_payable_fee_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 26, in test_payable_fee_no_crash
    |     payable_fee(s)
    |   File "D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7\daycare_fees.py", line 10, in payable_fee
    |     fee, discount = text.split('|')
    |                     ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7\test_daycare_fees.py", line 28, in test_payable_fee_no_crash
    |     assert False, f"payable_fee crashed with: {e!r}"
    | AssertionError: payable_fee crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_payable_fee_no_crash(
    |     s=None,  # or any other generated value
    | )
    +----------------------------------
=============================== short test summary info ===============================
FAILED test_daycare_fees.py::test_parse_enrolment_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_daycare_fees.py::test_hours_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_daycare_fees.py::test_payable_fee_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed, 3 passed in 13.27s ===============================
PS D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7>
```

SS3-FIXED PROGRAM OUTPUT:

```
PS D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7> py -3.11 daycare_fees.py
==== Running parse_enrolment ====
Enter age group:hours (e.g. toddler:25): toddler:25
Age Group and Hours: ('Toddler', 25.0)

==== Running hours_average ====
Enter weekly hours as CSV (e.g. 20,25,30): 25
Average Hours: 25.0

==== Running payable_fee ====
Enter fee|discount (e.g. 3000|250): 3000|250
Payable Fee: 3290.0
PS D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7>
```

SS4-PASSING HYPOTHESIS TESTS:

```
PS D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7> py -3.11 -m pytest test_daycare_fees.py
============================================================================= test session starts ==============================================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15, langsmith-0.14.16
collected 6 items


test_daycare_fees.py ......                                                                                                                  [100%]

============================================================================== 6 passed in 8.93s ===============================================
PS D:\L6_L7_48_daycare_fees\L6_L7_48_daycare_fees\Lab 7>
```